

HomeSenseKD: Heterogeneous Federated Knowledge Distillation for CSI-Based Indoor Sensing

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HomeSenseKD Research

<https://github.com/gadm21/HomeSenseKD>

Abstract—Indoor human activity recognition (HAR) using Wi-Fi Channel State Information (CSI) offers a non-intrusive, privacy-preserving alternative to camera-based sensing. Deploying such systems across multiple homes introduces three compounding challenges: statistical heterogeneity of activity distributions, architectural diversity of edge devices, and the prohibition on sharing raw sensor data. We present FedMKS (Federated Mixed Knowledge Sharing), a heterogeneous federated knowledge-distillation algorithm for CSI-based indoor sensing. FedMKS organises clients into architecture tiers and couples intra-tier weight sharing with cross-tier feature-carrier distillation on a public set, enabling model-heterogeneous collaboration without exchanging raw data or full model weights. We evaluate FedMKS against FedMD [17], FedAKD [17], [20], FedAvg [15], FedProx [16], and FedDF [21], together with Local and Centralised baselines, on the HomeOccupancy CSI dataset and two widely used image benchmarks (MNIST, CIFAR-10) under IID and non-IID partitions. On HomeOccupancy, FedMKS attains 90.4% validation accuracy under IID partitioning—within 0.5% of the centralised ceiling. Under non-IID label skew, soft-label-only distillation underperforms even all-small weight sharing, motivating FedMKS’s hybrid design. Code and datasets are publicly available.

I. INTRODUCTION

Wi-Fi Channel State Information (CSI) encodes fine-grained multipath propagation effects that shift measurably when a person moves, breathes, or changes posture. By analysing CSI time series from commodity access points, it is possible to recognise indoor activities without deploying cameras or wearables—an increasingly attractive proposition for smart homes and assisted living [1], [2].

Training accurate CSI-based models, however, requires labelled data that is inherently private and difficult to collect at scale. Federated Learning (FL) mitigates this by allowing devices to collaborate without sharing raw data; each client trains locally and contributes only model updates or soft predictions [15]. Beyond privacy, FL brings a natural benefit for sensing: a client whose environment lacks certain classes (e.g., no “smoking” behaviour observed) can still learn to recognise them through knowledge transferred from peers that have seen those classes.

A persistent practical obstacle is *model heterogeneity*: edge devices range from resource-constrained sensors to capable gateways, making a one-size-fits-all model architecture infeasible. Standard weight-sharing algorithms (FedAvg [15], FedProx [16]) require all clients to share the same architecture;

knowledge-distillation approaches (FedMD [17], FedAKD [20]) avoid weight exchange entirely but forego the benefits of model aggregation. FedMKS bridges this gap by organising clients into architecture tiers and simultaneously sharing weights within tiers and feature carriers across tiers, matching or exceeding both pure weight-sharing and pure KD baselines in semi-heterogeneous settings.

Contribution. We propose **FedMKS** (Federated Mixed Knowledge Sharing), a federated knowledge-distillation algorithm for model-heterogeneous CSI sensing. FedMKS (i) partitions clients into architecture tiers and shares classifier weights within each tier, (ii) shares distilled feature carriers across all tiers on a public set, and (iii) applies Mixup augmentation to the public set before distillation. A dual-head tiered model decouples private classification from public-set distillation, so clients of different capacity collaborate without exchanging raw data or full model weights.

The remainder of this paper is structured as follows. Section II surveys related work. Section III defines the system model. Section IV describes all algorithms and the shared model. Section V presents three-phase experimental design and results. Section VI concludes.

II. RELATED WORK

A. Wi-Fi Sensing

Wi-Fi CSI sensing has progressed from coarse-grained indoor localisation to fine-grained activity and gesture recognition. Ma et al. [1] provide a comprehensive survey of CSI-based sensing techniques, cataloguing feature extraction strategies from amplitude and phase of subcarrier signals. Yousefi et al. [2] demonstrate that statistical features of CSI effectively distinguish human activities without device-attached sensors. Adib and Katabi [3] pioneered through-wall motion detection using commodity 802.11 hardware, establishing that reflections of Wi-Fi signals encode human presence. Wang et al. [4] introduced E-Eyes, one of the first systems mapping CSI amplitude patterns to specific indoor activities such as walking and falling. Chen et al. [5] showed that STFT-based time-frequency features of CSI enable real-time gesture classification with high accuracy. Han et al. [6] applied deep learning to CSI streams for healthcare HAR, achieving strong accuracy on activities of daily living with convolutional architectures. Jiang et al. [7] addressed environment dependence

in CSI-based HAR, showing that domain-adaptation techniques improve cross-location generalisability. Liu et al. [8] proposed multi-task deep learning to jointly perform localisation and activity recognition from a single CSI stream.

B. Federated Learning for Sensing

Sozinov et al. [9] were among the first to apply federated machine learning to HAR, demonstrating that local models collaboratively achieve accuracy competitive with a centralised model while preserving privacy. Wu et al. [10] extended FL to CSI-based sensing, showing that the non-IID nature of per-home activity distributions degrades FedAvg and motivating personalised federated approaches. Ek et al. [11] applied FL to WiFi-based user authentication, demonstrating that even simple weight-averaging benefits from multi-device participation when data distributions are complementary. Ouyang et al. [12] proposed ClusterFL, which groups FL clients by activity-distribution similarity, reducing inter-client disagreement and improving convergence on heterogeneous sensing datasets. Presotto et al. [13] introduced semantic-aware aggregation in FL for HAR, weighting client contributions by model confidence rather than dataset size. Zhao et al. [14] showed that FL accuracy degrades sharply when data are non-IID, especially for classification tasks, motivating specialised aggregation and distribution strategies.

C. Federated Learning with Knowledge Distillation

McMahan et al. [15] introduced FedAvg, establishing the standard of aggregating locally trained weights at a central server and demonstrating strong performance on IID splits. Li et al. [16] extended FedAvg with a proximal regularisation term (FedProx) that stabilises training under system and statistical heterogeneity. Li and Wang [17] proposed FedMD, in which fully heterogeneous clients share only soft predictions on a public dataset, eliminating any constraint on client architecture at the cost of weaker gradient signal. Hinton et al. [18] established the foundational knowledge distillation framework—training a student to match teacher soft outputs—on which FedMD, FedAKD, and FedMKS build. Shen et al. [19] proposed federated mutual learning, enabling peer-to-peer distillation between clients without a public dataset. Lin et al. [21] proposed **FedDF**, in which the server collects client model outputs on a shared public dataset, constructs an ensemble prediction, and trains a fused student model via server-side knowledge distillation, improving accuracy and robustness compared to naive averaging. Unlike FedDF, which concentrates all distillation on the server, FedMKS distributes distillation across clients, substantially reducing server computation at the cost of higher client workload and communication overhead.

III. SYSTEM MODEL

A. Notation

B. Network and Data Model

Consider a star-topology FL network with a central server and $K=20$ clients. Each client $k \in \mathcal{K}$ holds a private labelled

TABLE I
SYMBOL TABLE.

Symbol	Meaning
K	Number of clients ($K=20$)
$\mathcal{K}$	Client index set $\{1, \dots, K\}$
R	Number of FL communication rounds ($R=50$)
r	Round index, $r \in \{1, \dots, R\}$
$\mathcal{D}_k$	Private labelled dataset of client k
n_k	$ \mathcal{D}_k $; $n = \sum_k n_k$
$\mathcal{Y}$	Global label space; $C= \mathcal{Y} $
$\mathcal{Y}_k$	Local label subset, $\mathcal{Y}_k \subseteq \mathcal{Y}$
$\mathcal{D}^{\text{pub}}$	Shared public (unlabelled) dataset
t_k	Tier of client k : S, M, or L
θ_k	All trainable parameters of model f_k
θ_k^{bb}	Backbone parameters (shared by classification and distillation heads)
$\mathbf{W}_t^{(r)}$	Server-side averaged classifier weights for tier t at round r
d	Distillation-head output dimension (feature carrier)
$s_k^{(r)}$	Feature carrier from client k at round r , $s_k^{(r)} \in \mathbb{R}^{ \mathcal{D}^{\text{pub}} \times d}$
$\bar{s}^{(r)}$	Aggregated global carrier at round r
T, B	CSI window length (1500) and STFT bins (8)

dataset $\mathcal{D}_k = \{(x_i^k, y_i^k)\}_{i=1}^{n_k}$, where $x_i^k \in \mathbb{R}^{T \times (4+B)}$ is a preprocessed CSI window and $y_i^k \in \mathcal{Y}_k \subseteq \mathcal{Y}$ is the activity label. Under *IID* partitioning, each client receives $n_c=30$ labelled examples per class. Under *non-IID* partitioning, class proportions are drawn from a Dirichlet distribution $\mathbf{p}_k \sim \text{Dir}(\alpha)$ with $\alpha=0.5$, creating highly skewed per-client class distributions.

All clients share access to a freely available *public* dataset $\mathcal{D}^{\text{pub}}$ (e.g. HomeHAR for HomeOccupancy clients) that carries no private labels and is used exclusively for knowledge transfer.

C. Heterogeneous Model Architecture

Each client is assigned a compute tier $t_k \in \{\text{S}, \text{M}, \text{L}\}$ reflecting its hardware budget. Every tier instantiates a *dual-head* model $f_k = (\phi_k, h_k^{\text{clf}}, h_k^{\text{dist}})$, where ϕ_k is a shared backbone parameterised by θ_k^{bb} and the two heads are trained with decoupled losses:

- **Classification head** h_k^{clf} : trained on private data with cross-entropy loss $\mathcal{L}_{\text{CE}}(\theta_k) = -\frac{1}{n_k} \sum_i \log p_k(y_i^k | x_i^k)$.
- **Distillation head** h_k^{dist} : produces a d -dimensional feature carrier $s_k^{(r)} = h_k^{\text{dist}}(\phi_k(\mathcal{D}^{\text{pub}})) \in \mathbb{R}^{|\mathcal{D}^{\text{pub}}| \times d}$, aligned to the global aggregate via $\mathcal{L}_{\text{KD}}(\theta_k) = \frac{1}{|\mathcal{D}^{\text{pub}}|} \|s_k^{(r)} - \bar{s}^{(r-1)}\|_F^2$.

The three tiers use *distinct* backbones of increasing capacity: **S** is a three-layer Conv1D stack with global average pooling; **M** is a two-layer Conv1D stack followed by a BiLSTM(128); **L** pairs a convolutional front-end with a BiLSTM and a Transformer encoder block.

D. Traditional FL Objective and Its Limitations

Standard FL (FedAvg [15]) seeks a single global model θ^* minimising the weighted empirical risk:

$$\theta^* = \arg \min_{\theta} \sum_{k=1}^K \frac{n_k}{n} \mathcal{L}_{\text{CE}}^k(\theta), \quad (1)$$

aggregating client updates each round as $\theta^{(r)} \leftarrow \sum_k (n_k/n) \theta_k^{(r)}$. This formulation requires *all clients to share*

the same architecture, making it inapplicable when $t_j \neq t_k$. FedProx [16] adds a proximal term $\frac{\mu}{2} \|\theta_k - \theta^{(r-1)}\|^2$ to each client's loss to bound client drift under heterogeneous data, but still enforces architectural homogeneity.

E. Knowledge-Distillation-Based FL Objective

To support fully heterogeneous architectures, FedMD [17] replaces weight aggregation with shared-representation aggregation. The server maintains a global carrier matrix $\bar{s}^{(r)} = \frac{1}{K} \sum_k s_k^{(r)}$ and each client minimises the combined objective:

$$\mathcal{L}_k(\theta_k) = \underbrace{\mathcal{L}_{\text{CE}}(\theta_k; \mathcal{D}_k)}_{\text{private classification}} + \lambda \underbrace{\mathcal{L}_{\text{KD}}(\theta_k; \mathcal{D}^{\text{pub}}, \bar{s}^{(r-1)})}_{\text{KD alignment}}, \quad (2)$$

where λ balances local accuracy with global knowledge absorption. Because the carrier $\bar{s}^{(r)}$ lives in a *shared d -dimensional feature space*, clients with entirely different backbone architectures can participate in the same aggregation round.

F. Problem Definition

Goal. Given K clients with heterogeneous private datasets $\{\mathcal{D}_k\}$ and heterogeneous model tiers $\{t_k\}$, find per-client classifiers $\{f_k^{\text{clf}}\}$ that jointly maximise mean validation accuracy across clients, subject to three constraints:

- 1) *Privacy*: $\mathcal{D}_k$ never leaves client k ; only $s_k^{(r)}$ and/or θ_k^{bb} are transmitted.
- 2) *Architectural heterogeneity*: no direct weight transfer between clients of different tiers ($t_j \neq t_k$).
- 3) *Communication bound*: per-round client payload is the sum of a within-tier weight term and a cross-tier knowledge term. Pure weight-sharing methods transfer $\mathcal{O}(|\theta|)$, pure KD methods transfer $\mathcal{O}(|\mathcal{D}^{\text{pub}}| \cdot d)$, and FedMKS transfers both within-tier classifier weights $\mathcal{O}(|\theta_k|)$ and cross-tier feature carriers $\mathcal{O}(|\mathcal{D}^{\text{pub}}| \cdot d)$, but never full weights across tiers. Every client receives its tier's aggregated weights *and* the global carrier; it uses the carrier as targets to train on $\mathcal{D}^{\text{pub}}$.

FedMKS (Section IV) addresses all three constraints simultaneously by combining within-tier weight averaging (Eq. 1 applied per tier) with cross-tier feature-carrier distillation (Eq. 2), bridging the gap between FedAvg efficiency and FedMD flexibility.

IV. METHODS

A. Tiered Dual-Head Architecture

The three tiers use distinct backbones of increasing capacity: **Small**—Conv1D(128,3) $\rightarrow$ MaxPool $\rightarrow$ Conv1D(256,3) $\rightarrow$ MaxPool $\rightarrow$ Conv1D(256,3) $\rightarrow$ GAP $\rightarrow$ Dense(256); **Medium**—Conv1D(64,5) $\rightarrow$ MaxPool $\rightarrow$ Conv1D(128,3) $\rightarrow$ MaxPool $\rightarrow$ BiLSTM(128); **Large**—Conv1D(64,5) $\rightarrow$ MaxPool $\rightarrow$ Conv1D(128,3) $\rightarrow$ MaxPool $\rightarrow$ BiLSTM(128) $\rightarrow$ Transformer block (MHA-4-heads, FFN-256, LayerNorm) $\rightarrow$ GAP. The *classification head* (Dense $\rightarrow$ Softmax, CE loss) is trained on $\mathcal{D}_k$; the *distillation head* (Dense $\rightarrow$ ReLU with $d=64$ output units, MSE loss on $\bar{s}$) is trained on $\mathcal{D}^{\text{pub}}$. Gradient

Algorithm 1 FedMKS — one communication round

Require: Clients $\mathcal{K}$; tiers $t \in \{\text{S,M,L}\}$; $\mathcal{D}^{\text{pub}}$; E_{kd} , E_{loc} ; round r ; current global carrier $\bar{s}^{(r-1)}$ and tier weights $\{\mathbf{W}_t^{(r-1)}\}$

Ensure: Updated global carrier $\bar{s}^{(r)}$ and tier weights $\{\mathbf{W}_t^{(r)}\}$

- 1: **if** $r = 1$ **then**
- 2: **Server:** send empty payload to each k $\triangleright$ bootstrap
- 3: **else**
- 4: **Server:** broadcast $\bar{s}^{(r-1)}$ and $\mathbf{W}_{t_k}^{(r-1)}$ to each k
- 5: **end if**
- 6: **for** each client k **in parallel do**
- 7: **if** $r > 1$ and $\mathbf{W}_{t_k}^{(r-1)}$ received **then**
- 8: Load classifier weights $\leftarrow \mathbf{W}_{t_k}^{(r-1)}$
- 9: Receive carrier $\bar{s}^{(r-1)}$ into public-set targets
- 10: Fine-tune dist. head on $\mathcal{D}^{\text{pub}}$ for E_{kd} epochs minimising $\mathcal{L}_{\text{MSE}}(f_k^{\text{dist}}(\cdot), \bar{s}^{(r-1)})$
- 11: **end if**
- 12: Train clf. head on private data $\mathcal{D}_k$ for E_{loc} epochs minimising $\mathcal{L}_{\text{CE}}$
- 13: Return $s_k^{(r)} \leftarrow f_k^{\text{dist}}(\mathcal{D}^{\text{pub}})$ and $w_k^{(r)} \leftarrow \theta_k$
- 14: **end for**
- 15: **Server:** $\bar{s}^{(r)} \leftarrow \frac{1}{K} \sum_{k=1}^K s_k^{(r)}$
- 16: **Server:** $\mathbf{W}_t^{(r)} \leftarrow \text{FedAvg}(\{w_k^{(r)} : t_k=t\})$ for each tier t

flows are fully decoupled between heads; only the backbone is shared.

B. FedMKS: Mixed Knowledge Sharing

FedMKS (Fig. 1) is the proposed algorithm that addresses the limitations of both FedAvg [15] and FedMD [17] in heterogeneous networks. FedAvg requires all clients to share a single architecture, so it must use the smallest model in a heterogeneous network—sacrificing accuracy for capable clients. FedMD shares only feature carriers, accommodating full heterogeneity, but misses the accuracy gains of direct weight sharing. FedMKS simultaneously: (i) shares weights within clients of the same tier (intra-tier FedAvg); (ii) distills knowledge across all tiers via aggregated feature carriers on the public set; and (iii) applies Mixup augmentation to the public set before distillation (following FedAKD [20]).

In round $r=1$ the server holds no global state, so each client trains only on its private data and then uploads its initial feature carrier and classifier weights. From round $r=2$ onward, the server sends every client its tier's averaged classifier weights $\mathbf{W}_{t_k}^{(r-1)}$ and the global carrier $\bar{s}^{(r-1)}$; the client loads the weights, fine-tunes the distillation head on the public set using $\bar{s}^{(r-1)}$ as targets, and then trains the classification head on its private data for E_{loc} epochs. It returns the new carrier $s_k^{(r)}$ and classifier weights $w_k^{(r)}$ to the server.

Algorithm 1 formalises one FedMKS round.

Comparison with FedDF. FedDF [21] addresses the same heterogeneous FL problem via server-side ensemble distillation: the server runs all client models on $\mathcal{D}^{\text{pub}}$, forms an ensemble $\bar{s}^{(r)} = \sum_k \beta_k f(\mathcal{D}^{\text{pub}}, \theta_k)$, and trains a fused student by minimising $D_{\text{KL}}(\bar{s}_\tau \| f_\tau(\mathcal{D}^{\text{pub}}, \theta))$. This centralises both inference and gradient computation on the server, incurring $\mathcal{O}(KPF + E_{\text{kd}}P(\mathcal{F} + \mathcal{B}))$ operations per round, where $P = |\mathcal{D}^{\text{pub}}|$ and $\mathcal{F}, \mathcal{B}$ denote forward- and backward-pass costs.

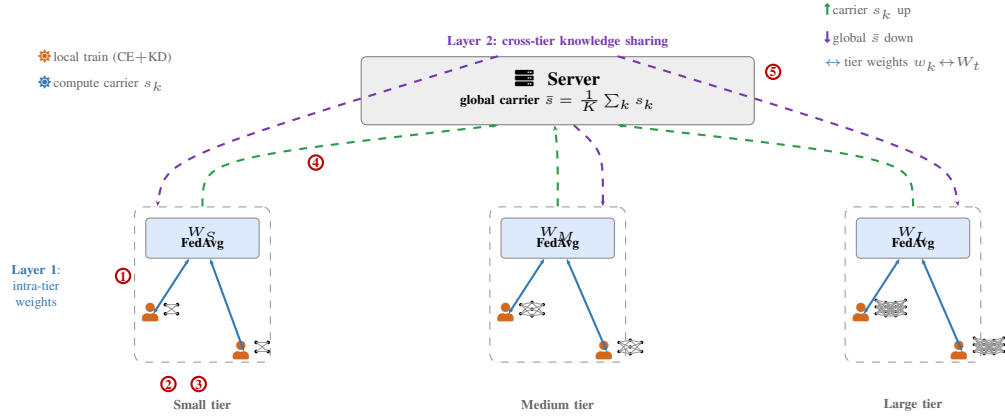


Fig. 1. **FedMKS overview.** Clients are grouped into three architecture tiers (Small, Medium, Large). Two cooperation layers operate each round: **Layer 1** (intra-tier) shares *model weights* — clients in a tier FedAvg their weights into a tier model W_t ; **Layer 2** (cross-tier) shares *knowledge* — every client uploads a feature carrier s_k to the server, which returns the global carrier $\bar{s}$ used as a distillation target. Steps: ① local training (CE on private data + KD on the public set), ②③ compute the local carrier, ④ upload carrier + weights, ⑤ server aggregates carriers globally and weights per tier.

TABLE II
ALGORITHM SUMMARY (✓ = FEATURE USED; † PHASE 3 ONLY).

Algo.	Soft	Wts	Topology
FedMD	✓	—	Heterogeneous
FedAKD	✓	—	Heterogeneous
FedMKS	✓	✓	Semi-het.
FedDF†	✓	✓	Heterogeneous
FedAvg	—	✓	Homogeneous
FedProx	—	✓	Homogeneous
Local	—	—	Isolated
Central	—	—	Centralised

TABLE III
HYPERPARAMETER SETTINGS (ALL EXPERIMENTS).

Parameter	Value
Rounds R	50
Clients K	20
IID samples / class / client	30 (CSI) / 100 (image)
Non-IID Dirichlet α	0.5
Local epochs E_{loc}	4 (CSI) / 5 (image)
KD epochs E_{kd}	2 (CSI) / 3 (image)
Carrier dimension d	64
FedProx μ	0.01
Alignment samples	10 000 (cap)
CSI window T	1500
STFT bins B	8

FedMKS inverts this design: the server performs only tensor aggregation $O(\sum_t |\mathbf{W}_t| + KPd)$ with no neural-network passes, shifting the distillation burden to clients—making it far more suitable when the coordinator is a lightweight edge gateway.

Table II summarises all evaluated algorithms.

V. EXPERIMENTS AND RESULTS

A. Experiment Design

Datasets. *HomeOccupancy* [22]: 3-class CSI (empty/sleep/work), sessions 1–2 as train, session 3 as test; public data is the *HomeHAR* CSI corpus [22]. *MNIST*: 10-class digit images; public data is CIFAR-10. *CIFAR-10*: 10-class object images; public data is MNIST.

Hyperparameters (Table III): $R=50$ rounds, $K=20$ clients, $n_c=30$ samples/class/client (IID), $\alpha=0.5$ (Dirichlet non-IID), $E_{loc}=4$, $E_{kd}=2$, $\mu=0.01$ (FedProx), $n_{align}=10,000$, $T=1500$, $B=8$. Tier distribution (Phase 1): 35% S, 35% M, 30% L. The image benchmarks (MNIST, CIFAR-10) use a larger per-client budget reflecting their greater data availability— $n_c=100$ samples/class, $E_{loc}=5$, $E_{kd}=3$; all other settings are shared. Results are mean validation accuracy across up to 3 independent seeds; runs still in progress are single-seed (marked †).

Phase 1 evaluates all 7 algorithms on 3 datasets under uniform tier heterogeneity (35% S/35% M/30% L) for both IID and non-IID data partitions. Because weight aggregation requires identical architectures, FedAvg and FedProx degrade every client to the smallest (small) model, whereas the KD methods and FedMKS retain their assigned heterogeneous tiers. **Phase 2** ablates tier-heterogeneity distribution on HomeOccupancy for FedMD across three configurations: all_small (100% S), uniform (35/35/30), and skewed (60% S/30% M/10% L). **Phase 3** directly compares FedMKS and FedDF on HomeOccupancy under the IID partition (uniform tier distribution, single seed), reporting (i) validation accuracy, (ii) per-round server computation time, (iii) per-round mean client computation time, and (iv) total communication bytes per round.

B. Results

1) *Phase 1 — Algorithm Comparison:* Figs. 2 and 3 show validation accuracy curves over 50 federated rounds under IID and non-IID (Dirichlet $\alpha=0.5$) partitions, respectively. The HomeOccupancy non-IID condition is omitted while its runs complete.

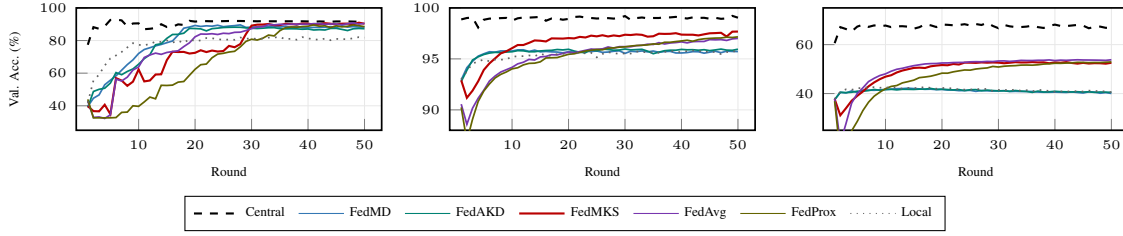


Fig. 2. Validation accuracy over 50 federated rounds under IID partitioning (mean across completed seeds). Panels (left-to-right): HomeOccupancy, MNIST, CIFAR-10. All algorithms use uniform tier distribution (35% S/35% M/30% L); FedAvg/FedProx are restricted to the all-small architecture. On HomeOccupancy, FedAvg and FedMKS match the centralised ceiling while soft-label-only methods lag slightly. On MNIST all FL methods approach centralised accuracy; on CIFAR-10 the cross-domain public pool suppresses all federated methods $>10\%$ below Central. † HomeOccupancy FedMD/FedAKD/FedMKS are single-seed; remaining seeds are in progress.

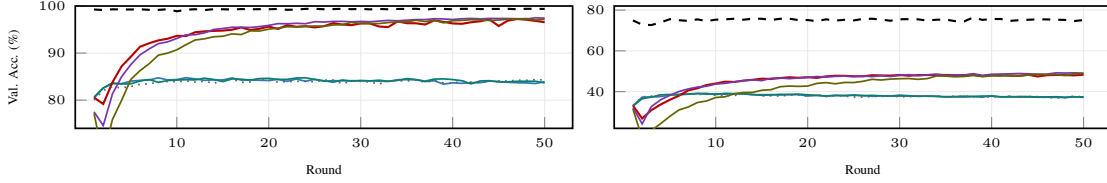


Fig. 3. Validation accuracy over 50 federated rounds under non-IID partitioning (Dirichlet $\alpha=0.5$; mean across completed seeds). Panels (left-to-right): MNIST, CIFAR-10. The HomeOccupancy non-IID condition is omitted while its runs complete. On MNIST, FedMKS/FedAvg/FedProx stay near-centralised while soft-label-only methods plateau at $\approx 84\%$. On CIFAR-10, FedAvg edges out FedProx and FedMKS.

HomeOccupancy. Under IID, FedMKS (90.4%) and FedAvg (89.5%) closely approach the centralised ceiling (90.8%), while FedMD (88.5%) and FedAKD (87.0%) lag due to the absence of weight sharing. Local training plateaus at 80.4%, confirming the value of collaboration. The non-IID evaluation of the soft-label methods is still in progress and is therefore omitted from Fig. 3; completed weight-sharing runs (FedAvg 38.4%, FedProx 33.7%) already trail local-only training (60.1%), consistent with the known difficulty of aggregation under Dirichlet skew.

MNIST. MNIST is an easier benchmark that confirms algorithm ordering under ideal conditions. IID: Central 99.0%, FedMKS 97.7%, FedAvg/FedProx 97.0%/97.2%, FedMD/FedAKD/Local all $\approx 95.8\%$. Non-IID: FedAvg 97.4%, FedProx 97.2%, FedMKS 96.6% remain competitive, while FedMD/FedAKD/Local plateau at $\approx 84\%$. Notably, the soft-label-only methods—despite retaining heterogeneous (larger) architectures—fall ≈ 13 points below FedAvg/FedProx, which are restricted to the smallest model.

CIFAR-10. CIFAR-10 reveals the fundamental difficulty of cross-dataset public/private knowledge transfer (public is MNIST, private is CIFAR-10). All FL methods fall 10–30% below centralised. FedAvg (53.8%) marginally leads FedProx (52.9%) and FedMKS (52.6%) under IID; under non-IID FedAvg (49.1%) and FedProx (48.9%) edge out FedMKS (48.3%). KD-only methods (FedMD 37.2%, FedAKD 37.4%) do not exceed local training (37.3%), suggesting MNIST soft labels provide negligible signal for CIFAR-10 learning.

a) Key observation.: Two non-IID results motivate FedMKS. First, weight sharing is a stronger signal than cross-architecture distillation alone: FedAvg/FedProx reach

$\approx 97\%$ on MNIST non-IID while FedMD/FedAKD plateau at $\approx 84\%$. Second—and surprisingly—this holds even though FedAvg/FedProx are forced onto the smallest architecture, so distillation by itself achieves *lower* accuracy than simply degrading every client to the weakest model. FedMKS is designed to recover model-sharing-level accuracy while preserving the heterogeneity that only distillation permits.

2) Ablation Studies:

a) Tier heterogeneity (Phase 2).: Phase 2 investigates how tier-heterogeneity distribution affects FedMD on HomeOccupancy, comparing *all_small*, *uniform*, and *skewed* configurations under both IID and non-IID settings (Fig. 4).

Under IID, tier composition has negligible effect: *all_small*, *uniform*, and *skewed* all converge to $\approx 86\%$, indicating that FedMD soft-label distillation is robust to model-capacity heterogeneity when per-client labels are balanced. Under non-IID the picture changes: *uniform* (61.5%) outperforms both *all_small* (59.1%) and *skewed* (58.6%). Because *all_small* (100% S) and *skewed* (60% S) concentrate clients on the weakest architecture, they converge below the balanced *uniform* mix—i.e. small-dominated tier distributions underperform heterogeneous ones under label skew.

b) FedMKS vs. FedDF (Phase 3).: Phase 3 compares FedMKS and FedDF [21] on HomeOccupancy under the IID setting (uniform tier distribution, single seed, 30 rounds, 10 clients), recording validation accuracy together with per-round server compute, client compute, and communication bytes (Fig. 5, Table IV). FedDF converges faster, plateauing near round 10, and holds a slightly higher converged accuracy

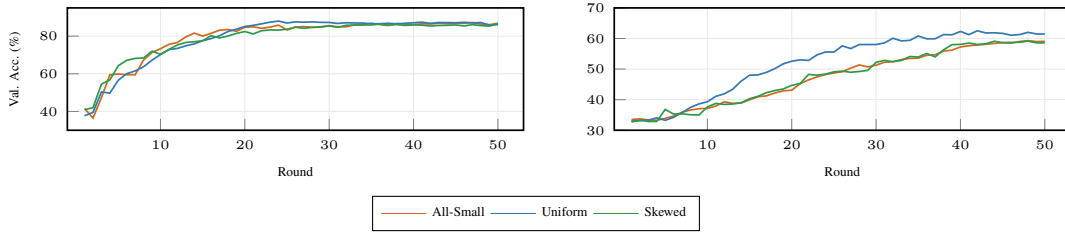


Fig. 4. FedMD tier-heterogeneity ablation on HomeOccupancy. Validation accuracy over 50 rounds under three tier distributions—all_small (100% S), uniform (35/35/30), skewed (60/30/10)—for IID (left) and non-IID (right) partitions. Tier composition has negligible effect under IID; under non-IID the small-dominated distributions (all_small, skewed) converge below the balanced uniform mix.

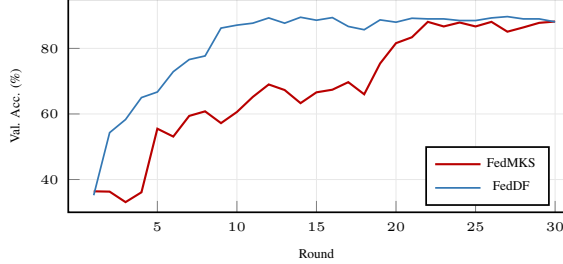


Fig. 5. Phase 3: mean client validation accuracy of FedMKS and FedDF on HomeOccupancy under IID (single seed, 30 rounds, 10 clients). FedDF plateaus earlier (round ≈ 10) near 89%; FedMKS climbs more gradually, reaching $\approx 88\%$ by round 30.

(89.0% vs. 87.1% mean over the final five rounds), though FedMKS is still climbing and edges ahead at the final round (88.2% vs. 88.1%). The efficiency profiles differ sharply. FedDF concentrates the distillation cost on the server— ≈ 243 s per round of server-side ensemble training—whereas FedMKS performs essentially no server compute (≈ 0.06 s per round, aggregation only) and instead distributes the distillation across clients, which run in parallel (≈ 93 s vs. ≈ 25 s per client per round). FedMKS thus removes the serial server bottleneck at the cost of higher per-client work and $\approx 22\%$ more communication (37.9 vs. 31.0 MB per round). The model-weight term is identical for both (31.0 MB): each client sends and receives its own tier’s weights, ≈ 1.2 – 2.2 MB depending on tier. FedMKS’s entire ≈ 6.9 MB overhead is the soft-label carrier exchange—each client uploads its carrier and downloads the global carrier ($|\mathcal{D}^{\text{pub}}| \times d$ each way), which FedDF avoids by distilling on the server.

VI. CONCLUSION

We presented **FedMKS**, a heterogeneous federated knowledge-distillation algorithm for CSI-based indoor activity sensing. The dual-head tiered model and the FedMKS algorithm demonstrate that combining intra-tier weight sharing with cross-tier feature-carrier distillation achieves near-centralised accuracy on HomeOccupancy ($\Delta \approx 0.5\%$ under IID) while preserving full architectural heterogeneity. Phase 1 results on three datasets expose a clear hierarchy: weight-sharing methods dominate under IID; under non-IID, soft-label-only distillation underperforms even all-small weight sharing, while

TABLE IV
PHASE 3 EFFICIENCY COMPARISON ON HOME OCCUPANCY (IID, 30 ROUNDS, 10 CLIENTS). ACCURACY IS THE MEAN OVER THE FINAL FIVE ROUNDS, AS IN PHASE 1. FEDMKS REMOVES THE SERVER-SIDE DISTILLATION BOTTLENECK BY DISTRIBUTING IT ACROSS PARALLEL CLIENTS, AT HIGHER PER-CLIENT COMPUTE AND MODESTLY MORE COMMUNICATION.

Metric (per round)	FedMKS	FedDF
Final val. accuracy (%)	87.1	89.0
Server compute (s)	0.06	243.2
Mean client compute (s)	93.1	24.6
Comm.: model weights (MB)	31.0	31.0
Comm.: soft-label carriers (MB)	6.9	0.0
Communication total (MB)	37.9	31.0
Total communication (MB, 30 r.)	1137.7	930.1

FedMKS’s hybrid of intra-tier weight sharing and cross-tier distillation recovers near-centralised accuracy. Phase 2 confirms that small-dominated tier distributions converge below a balanced heterogeneous mix under label skew. Phase 3 confirms that FedMKS attains comparable accuracy to FedDF (87.1% vs. 89.0%) while eliminating its ≈ 243 s/round server-side distillation bottleneck—FedMKS uses ≈ 0.06 s/round of server compute, distributing distillation across parallel clients at $\approx 22\%$ higher communication cost. The public code and datasets provide a reproducible foundation for future work on privacy-preserving sensing.

VII. LIMITATIONS AND FUTURE WORK

Single tier distribution in Phase 1. Phase 1 uses only uniform tier distribution (35/35/30). Varying heterogeneity as in Phase 2 would reveal whether FedMKS advantages are consistent across tier compositions.

Communication cost. FedMKS transmits both soft labels and per-tier weights, incurring higher communication overhead than FedMD. Compression (quantisation, top- k sparsification) of weight updates would improve deployment feasibility on bandwidth-constrained IoT networks.

Cross-domain public datasets. CIFAR-10 experiments show that a public dataset from a different domain (MNIST) provides little useful signal. Investigating domain-adapted public data, or constructing virtual public datasets from client gradients, remains an open problem.

Asynchronous FL. HomeSenseKD assumes synchronous rounds. Real deployments exhibit client dropouts and straggler effects; adapting FedMKS to asynchronous aggregation is an important practical extension.

CODE AND DATA AVAILABILITY

Source code is available at <https://github.com/gadm21/HomeSenseKD>. HomeOccupancy and HomeHAR datasets are hosted on HuggingFace at <https://huggingface.co/gadgadgad>.

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